Lantern Supply Co. Repair Center Bench Manual

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This manual is for the technicians at the Repair Center, 1180 Downsville Pike, Dock B, Hagerstown. It covers intake, the bench rules, and the service procedures for every current product and for the Beacon 400, which we discontinued in March 2025 and still repair while parts last. Read the whole manual before your first unsupervised bench day. If a procedure here disagrees with the product's own service bulletin, the bulletin wins; tell Walt so the manual gets fixed.

Intake

Every lantern that comes through Dock B gets a repair ticket before anyone touches it. Ticket numbers are RT- followed by five digits (for example RT-20931) and are issued by the intake terminal. The ticket records the customer's name, the order number if there is one, the model, the serial number from the base plate, the fault as the customer described it, and the condition on arrival, including scratches, dents and missing parts. Photograph every lantern from the front and from the base before you open the box flaps the rest of the way.

Each ticket gets a coloured tag tied to the bail or handle:

- Red tag: warranty repair. The claim already has a WC- number from Customer Service. No charge to the customer, and the return shipping is on us.
- Yellow tag: paid repair. The customer has agreed to the bench fee and to an estimate for parts.
- Green tag: Second Light refurbishment. These are returns and trade-ins being rebuilt for the refurbished shelf. There is no customer waiting, so they are worked after red and yellow tags.

The bench fee for a paid repair is $25.00 plus parts at the catalogue price of the part. The fee is waived under warranty. If a yellow-tag repair turns out to need parts beyond the estimate by more than $10, stop and have Customer Service call the customer before you continue. Do not guess at what the customer would want.

Turnaround for a paid repair is 10 business days from the day the lantern arrives at Dock B, counting the arrival day as day zero. Warranty repairs are worked in the same queue and carry the same target. If a part is on backorder, the ticket is flagged "waiting on parts" and the customer is emailed on the third day of waiting and every five business days after that.

Fuel lanterns arrive with fuel in them more often than you would think, whatever the packing instructions say. Drain any fuel lantern over the drain tray at the intake bench before it goes to a shelf, and log the drained fuel in the fuel room book. A lantern that smells of kerosene when the ticket says white gas gets a note on the ticket; it changes what you look for.

Bench rules

1. Every repaired lantern runs for 30 minutes on the bench before it goes back to the customer. LED lanterns run on high for the first 10 minutes and low for the rest. Fuel lanterns run on high for the whole 30 minutes on the vented bench, bench 3 or bench 4 only. Record the start and end time on the ticket. A lantern that fails the burn-in goes back to the start of the procedure, not to the shipping bench.
2. Fuel is lit only at benches 3 and 4, under the hood, with the extractor running. The extractor switch is on the pillar between the two benches; the indicator must be green.
3. No fuel is stored at the benches. Fill from the one-quart bench can and return the can to the fuel room cabinet at the end of the shift.
4. Lithium packs are stored in the steel cabinet marked LI, never on the bench overnight. A swollen pack goes straight into a sand bucket and is logged for the hazardous waste pickup.
5. Wear safety glasses whenever a globe, chimney or mantle is being handled. Mantles are brittle once burned in; the ash is fine enough to get into eyes.
6. Torque values in this manual are limits, not targets, unless the step says otherwise. We have broken more generator fittings by over-tightening than we have ever lost to leaks.
7. Clean the bench and empty the drain tray at the end of each shift. The last technician out checks the extractor is off and the fuel cabinet is locked.
8. If you are not sure, stop and ask. A delayed repair costs us a day; a lantern that comes back twice costs us the customer.

Northstar Dual-Fuel (LS-ND2)

The Northstar is our most repaired product by a wide margin, which is a function of how many we sell and of how it is used, not of any weakness. Nearly every Northstar fault is one of four things: a clogged generator, a dry pump cup, a torn mantle, or the wrong fuel. Ask about the fuel first. The Northstar burns Clearburn white gas or unleaded gasoline. It does not burn kerosene, and a tank that has had kerosene in it will have a varnished generator and a sticky check valve. That is a paid repair even inside the warranty period.

Tools for the Northstar bench: 11 mm and 13 mm open-ended wrenches, the generator wrench (part T-ND2, the thin 9 mm one with the bent handle), a small flat screwdriver, the pump cup oil bottle, a nylon brush, and the torque driver set to 2.5 N·m for checking.

Generator tube replacement

The generator (part G-GEN2) is the tube that runs up through the burner and carries fuel from the tank to the mantles. Signs of a bad generator are a weak yellow flame that will not steady, pulsing light, and a tank that holds pressure but will not deliver it. The part is $13.00, so a typical generator replacement on a yellow tag comes to $38.00 with the bench fee.

1. Confirm the tank is drained and depressurised. Open the filler cap slowly over the tray.
2. Lift off the ventilator and the globe (part G-ND2). Set the globe on the padded shelf, not on the bench.
3. Turn the control valve to the fully open position so the cleaning needle is clear of the generator tip.
4. Loosen the generator nut at the base of the burner with the 11 mm wrench, then the fitting at the valve body with the generator wrench. Draw the old generator straight up and out.
5. Fit the new G-GEN2 with the tip toward the burner. Start both fittings by hand.
6. Tighten the valve body fitting hand-tight plus a quarter turn. Never more than 2.5 N·m. Check with the torque driver if you are new to this; the fitting is brass and strips easily.
7. Tighten the generator nut the same way.
8. Pressurise with 20 strokes of the pump, no fuel in the tank, and soap-test both fittings. Any bubbles, back off and reseat. Do not tighten further.
9. Fill with Clearburn white gas, fit the mantles, light under the hood and run the 30-minute burn-in.

Pump cup service

The pump cup is the leather washer on the end of the plunger. When it is dry it will not seal and the customer reports "the pump does nothing". Oil the pump cup with 3 drops of Clearburn pump oil every 10 fills; tell the customer this on the return note, because it is the one piece of maintenance that stops most Northstars coming back.

1. Unscrew the pump cap a quarter turn and withdraw the plunger.
2. Inspect the cup. A cup that is cracked or glazed hard is replaced (part P-CUP2); one that is merely dry is oiled.
3. Put 3 drops of pump oil on the cup and work it in with your thumb until it is pliable.
4. Refit and pump 10 strokes. The resistance should build by the fourth stroke.

Mantle replacement

Mantles are consumables and are not a warranty item. The Northstar takes two mantles, part M-21, sold as a 2-pack. A mantle with a hole, a tear or a dark patch is replaced; one that is simply grey is fine.

1. Remove the globe and brush the old mantle ash off the burner caps with the nylon brush.
2. Tie the new mantle to the burner cap with the string that comes on it and trim the string to 10 mm.
3. Light the mantle with a match from the bottom and let it burn to ash before you pressurise the tank. Do not touch it once it is burned in.
4. Refit the globe and check it seats on the gasket all the way round.

Globe

The Northstar globe is part G-ND2. A cracked globe is replaced; a smoked one is cleaned with the glass cleaner and a paper towel, never with anything abrasive. Fit the globe with the Lantern Supply mark facing the control valve so the customer can read the dial through it.

Beacon 600 (LS-600)

The Beacon 600 runs on four D cells and almost never has an electrical fault beyond corroded contacts. The faults we see are a cracked globe, a globe gasket that has let water in, and contacts corroded by a cell left in too long. Tools: a Phillips #2, contact cleaner, the fibre contact brush, and the gasket pick.

Globe and gasket

1. Unscrew the top cap (counter-clockwise, four full turns) and lift the globe (part G-600) straight up.
2. The globe gasket is part G-600G. Lift it out with the pick, not with a screwdriver. Replace it if it is flattened, cracked, or if there was water in the base.
3. Seat the new G-600G in the channel with the lip facing up. Press it down all the way round.
4. Fit the globe and screw the cap down until it stops, then back it off a quarter turn so the gasket is not crushed.
5. Water test: run the shower head over the lantern for 2 minutes on the wet bench, then open it and check the base is dry. The Beacon 600 is rated IPX5 and a repaired one must meet it.

Battery contacts

1. Remove the base plate (three Phillips screws, 0.6 N·m on refit).
2. If a cell has leaked, neutralise the deposit with the vinegar swab, then rinse with the contact cleaner and dry.
3. Brush the four contacts and the two springs with the fibre brush until they are bright.
4. Fit four fresh test cells from the bench stock, check the lantern lights on both levels and in red night mode, then remove the test cells. We do not ship the Beacon 600 with cells.

Sentinel 1000 and 1000 Pro

The Sentinel 1000 (LS-1000) and Sentinel 1000 Pro (LS-1000P) share one body and one service procedure; the Pro has the larger pack, four light panels instead of two, and the solar top. They carry a 5-year warranty, so most Sentinels on the bench are red tags. The faults are a pack that will not hold charge, a panel latch that will not hold a panel, and the charge port.

Tools: a T8 Torx driver, the plastic spudger, the pack tester, and the latch spring tool.

Battery pack

The Sentinel 1000 battery pack is part B-1000-21 and is held by four T8 Torx screws. The Pro's pack is part B-1000P-21, same mounting. A pack tests bad if it holds less than 70% of its rated capacity after a full charge on the pack tester, or if it is swollen at all.

1. Discharge the lantern to below 20% on low before opening it. The charge indicator shows one bar.

2. Open the base: the four T8 Torx screws are under the rubber feet. Lift the feet with the spudger and keep them; they are not sold separately.
3. Lift the pack out by the pull tab and unplug the three-pin connector. Do not pull on the wires.
4. Fit the new B-1000-21 (or B-1000P-21), connect, and run the self-test by holding the power button for 8 seconds. Four blinks is a pass.
5. Refit the screws to 0.5 N·m and press the feet back on.
6. A bad pack goes to the LI cabinet with the ticket number written on it in marker.

Panel latch

1. Each detachable panel is held by a latch with a spring (part L-1000S). A panel that drops when the lantern is tilted has a tired spring.
2. Remove the panel. Press the latch tab in with the spring tool and lift the latch out.
3. Fit the new spring with the long leg toward the body, press the latch back until it clicks.
4. Fit the panel, invert the lantern and shake it gently. The panel must stay put.

Charge port

The USB-C charge port is on a small board (part PCB-1000C) and is replaced as a unit. A port that charges only when the cable is held at an angle is replaced, not re-soldered. The board is held by two T8 screws and a ribbon cable. Handle the ribbon with the spudger only.

Hagerstown Kerosene Lantern (LS-HK12)

The Hagerstown is a plain 12-inch hurricane lantern and there is not much to go wrong with it. It burns K-1 kerosene or Clearburn lamp oil and uses a #2 flat wick, part W-78, sold in a 5-pack. Faults are a smoky flame (wick), a flame that will not stay lit (wick or air tubes), and a tank that weeps at the seam (replace the lantern; we do not repair tanks).

Wick trimming and replacement

1. Raise the globe with the lever and lift the burner cone off.
2. Wind the wick up 10 mm. Trim it straight across with the wick scissors, then take the two corners off at 45 degrees. A square-cut wick smokes at the corners.
3. If the wick is charred for more than 25 mm, or is less than 100 mm long, replace it. Feed the new W-78 down through the burner with the raised side toward the winder knob.
4. Soak a new wick in lamp oil for 10 minutes before lighting it.

5. Light under the hood and set the flame so that it is just below the point where it smokes. Run the 30-minute burn-in.

Air tubes and burner

The two side tubes carry air to the burner. If they are dented so that a 4 mm rod will not pass, straighten them with the tube mandrel or replace the lantern. The burner itself (part BR-12) is replaced only when the winder gear strips, which happens about twice a year across the whole fleet.

Firefly 220 and 220R

The Firefly 220 (LS-220) runs on three AA cells; the Firefly 220R (LS-220R) is the rechargeable model with the 2,600 mAh pack and a USB-C port. They look the same from the outside except for the port cover, and the 220R weighs 30 g more. Check the model number on the base plate before you order parts: the switches are different and the 220's switch will not fit the 220R.

Switch

1. Remove the base (two Phillips screws on the 220, three on the 220R).
2. The switch is a tactile switch on the main board. Switch part numbers: SW-220 for the 220, SW-220R for the 220R. Both are soldered; this is the one soldering job on the Firefly bench.
3. Replace the switch, refit, and check both brightness levels.

Firefly 220R USB-C port

A 220R that will not charge is usually the cable, then the port, then the pack, in that order. Test with the bench cable (part C-15) before opening anything.

1. Open the base (three screws) and lift out the pack. Pack part is B-220R.
2. The port is part of the main board (PCB-220R); the board is replaced as a unit, $9.00 on a paid repair.
3. On refit, check the port cover seals; the 220R is IPX4 and the cover is part of that rating.
4. Charge to full on the bench (about 3.5 hours) and check the power bank output with the load tester: it should hold 5 V at 1 A for 10 minutes.

Heritage No. 7 (LS-H7)

The Heritage No. 7 is a solid brass oil lamp with a 10-year warranty, so treat every one as a red tag unless the ticket says otherwise. It burns Clearburn lamp oil only. Faults are a cracked chimney (part CH-7), a wick that will not wind, and tarnish that the customer wants removed. Tarnish is not a defect, but we polish the lamp as a courtesy on any repair.

Chimney

1. Lift the chimney straight up out of the gallery. Do not twist it; the gallery prongs will scratch the glass.
2. Fit the new CH-7 with the seam at the back, and check the four prongs hold it without play.

Wick

1. The Heritage takes a 7/8-inch flat wick, part W-H7. Wind the old wick up and out.
2. Feed the new wick with the weave running lengthwise. Trim it level with the top of the burner, then round the corners.
3. Soak it for 10 minutes in lamp oil before lighting.

Brass care

1. Polish with the brass polish in the Heritage drawer and a soft cloth, in straight strokes.
2. Never use the polish on the burner threads; wipe them with a dry cloth only.
3. A customer who asks how to keep the lamp bright is told: a soft cloth after each use, polish twice a year, and no lacquer.

Beacon 400 (LS-400), discontinued

The Beacon 400 was discontinued in March 2025. We stock its parts until March 2028 and repair it on yellow and red tags until then. The procedure is the same as the Beacon 600's; the globe is part G-400 and the gasket G-400G. There is no red night mode to test.

Parts bins

Parts are in the numbered bins along the north wall. The bin number is on the ticket when the part is picked. Catalogue prices apply on paid repairs.

Part	Fits	Bin	Price
G-GEN2 generator	Northstar LS-ND2	1	$13.00
P-CUP2 pump cup	Northstar LS-ND2	1	$4.50
M-21 mantle 2-pack	Northstar LS-ND2	2	$6.99
G-ND2 globe	Northstar LS-ND2	3	$18.99
G-600 globe	Beacon 600	4	$14.99
G-600G globe gasket	Beacon 600	4	$2.50
G-400 globe	Beacon 400	5	$12.99
G-400G globe gasket	Beacon 400	5	$2.50
B-1000-21 battery pack	Sentinel 1000	6 (LI cabinet)	$39.00
B-1000P-21 battery pack	Sentinel 1000 Pro	6 (LI cabinet)	$64.00
L-1000S latch spring	Sentinel 1000, 1000 Pro	7	$1.25
PCB-1000C charge board	Sentinel 1000, 1000 Pro	7	$11.00
W-78 #2 flat wick 5-pack	Hagerstown LS-HK12	8	$4.99
BR-12 burner	Hagerstown LS-HK12	8	$8.50
SW-220 switch	Firefly 220	9	$2.00
SW-220R switch	Firefly 220R	9	$2.00
B-220R battery pack	Firefly 220R	6 (LI cabinet)	$12.00
PCB-220R main board	Firefly 220R	9	$9.00
C-15 USB-C cable	Firefly 220R	9	$5.99
CH-7 chimney	Heritage No. 7	10	$21.99
W-H7 7/8-inch wick	Heritage No. 7	10	$3.50

Bin counts are taken on the first Monday of each month by whoever has the earliest shift. Any part below five in stock is written on the whiteboard and ordered that day.